

U.S. Multinational Investments and Copyright Protection

Walter G. Park

American University

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The statistical analysis of firm level data on U.S. multinational companies and their foreign affiliates was conducted at the Bureau of Economic Analysis, U.S. Department of Commerce, under arrangements that maintain legal confidentiality requirements. Views expressed in this paper are those of the authors and do not necessarily reflect the official positions of the U.S. Department of Commerce

Copyrights 101

Knowledge-Based Economy, International Trade, and Development:

- Disproportionate work in economics on the *Patent System*
- Art Works, Films, Books, Sound Recordings contribute to the stock of knowledge
- Software, Databases, Journals used in scientific research, R&D
- Copyright-Based Industries account for a significant share of GDP, trade, employment
 - World Intellectual Property Office (WIPO) studies on the contribution of copyright-based industries
 - Caution: Non-causal analyses
- Copyright-related industries account for the majority of U.S. outward global licensing
 - In 2015, shares were approximately: computer software 30%, audio-visual & related 17%, industrial processes 37%, and trademarks 12.0%

Copyrights 101

Motivation:

	FDI Stock (Non-Bank)		Affiliated R&D		Licensing Receipts	
	2006	2015	2006	2014	2006	2015
Billions (real 2005 dollars)	1011	1376	29	44	81	104
Share of Manufacturing	0.42	0.38	0.82	0.62	0.39	0.37
Share of Copyright Industries	0.10	0.11	0.04	0.08	0.45	0.47
Percent in non-G6 countries	47%	69%	63%	65%	46%	66%
<i>Share from unaffiliated parties</i>					0.34	0.38

G6 countries are: Canada, France, Germany, Italy, Japan, and the United Kingdom

Manufacturing licensing is measured by the licensing of *Industrial Processes*

Background

Copyrights in Brief:

- Protect Creative Works
- Protect Expression, not Ideas
- Automatic Protection
 - To waive specific rights, go to *Creative Commons*
- Registration is not mandatory
- Copyright Systems are founded upon a balance
 - Rights versus Exceptions

State of Research

Basic Theory (Watt (2000), Landes & Posner (2003), Maskus (2012)):

- Stronger copyright protection and enforcement increases cost of reproduction, reduces imitation, increases return to rights holder, and generates incentives for creation and product launches
- But it increases costs of access and creates transactions costs (burden of clearing rights and/or working around existing rights, especially if limited modes of expression exist)

Empirical Evidence (few):

- Single country (US or Canada)
 - Baker and Cunningham (2006), Ku et al. (2009)
- Single product (books, music, film)
 - Liebowitz (2008), Heald (2008), Png and Wang (2009), Li et al. (2013), Giorcelli and Moser (2015)
- Piracy, File-sharing studies find crowding out is not 1:1
 - Rob and Waldfogel (2006), Oberholzer-Gee and Strumpf (2007), Danaher et al. (2013)
- Limited investigations beyond copyright industry
 - P. Smith et al. (2009) study effect on relevant Infrastructure
 - Lerner (2011) studies effect on cloud computing

Contributions of Paper

- Explains how copyright can affect *technological development*
- Shows that the effects of copyright reform on technology transfers vary by industry characteristics (esp. prior usage/reliance on copyright protection).
- Broadens the scope of the potential impact of copyright systems beyond the traditional copyright sector, such as incentives for investment in complementary industries.
 - Demand Spillovers: Expansion of Copyright Industries provides incentives for firms in complementary industries to develop or invest in inputs and products related to the production or consumption of creative works.
- This current study focuses on international evidence, using U.S. multinational firm-level data:
 - Variation in copyright systems is greatest across countries
 - U.S. companies are the leading holders of copyrighted works
 - Markets for creative works have expanded across the world
 - Copyright reforms have occurred under international agreements and are coordinated inter-governmentally
 - Intellectual property concerns arise especially in developing countries
 - Access to creative works is important for global economic development

Estimating Equations

- How dependent are U.S. copyright firms' FDI, international licensing, and research on **copyright regimes** abroad?

$$\ln(TECH_{int}) = \alpha_i + \alpha_n + \alpha_t + \beta_1 \ln(C_{nt}) + \beta_2 \ln(C_{nt})R_j + \gamma \mathbf{X}_{int} + \varepsilon_{int}$$

- Do expansions in foreign **copyright markets** influence investments by firms in the complementary industries?

$$\ln(TECH_{int}) = \alpha_i + \alpha_n + \alpha_t + \beta_1 \ln(M_{nt}) + \beta_2 \ln(M_{nt})R_j + \gamma \mathbf{X}_{int} + \varepsilon_{int}$$

where TECH denotes technology transfers, C copyright protection, M copyright market size, and X a vector of control variables. R denotes a conditioning factor (e.g., market *rank* based on an industry's overall domestic or global sales during the sample period, normalized by median sales). i indexes firm, t time, n country, and j industry.

U.S. MNCs

Seventy countries

Annual Data: 1989 - 2011

Firms in 2 Broad Industries:

④ Copyright-Related Industries

- 212 U.S. Parent Companies

- 182 of them have presence in G6 countries (Can, Fra, Ger, Ita, Jpn, UK)
- 136 of them have presence in non-G6 countries (64 other countries)

② Complementary Industries

- 515 U.S. Parent Companies

- 436 of them have presence in G6 countries (Can, Fra, Ger, Ita, Jpn, UK)
- 354 of them have presence in non-G6 countries (64 other countries)

Source: BEA

- *Annual Surveys and Benchmark Surveys of USDIA*
- *Quarterly Balance of Payment Surveys of USDIA*
- *Quarterly Survey of Transactions in Selected Services and Intellectual Property with Foreign Persons*
- *Annual Survey of Royalties, Licensing Fees, and Other Receipts and Payments for Intangible Rights between U.S. and Unaffiliated Foreign Persons*

Industry Coverage and Copyright Sales

NAICS code	Copyright-related Industries	Domestic Market Sales Rank
5111	Newspaper, periodicals, books, and directories	5
5112	Software publishers	1
5121	Motion picture and video industries	10
5122	Sound recording industries	11
5151	Radio and television broadcasting	4
5152	Cable and other subscription programming	3
5161	Internet Publishers	16
5171	Wired telecommunications carriers	13
5172	Wireless telecommunications (excluding satellite)	7
5173	Telecommunications Re-sellers	12
5174	Satellite Telecommunications	14
5175	Cable Distribution	2
5179	Other Telecommunications	9
5181	Internet Service Providers, Web Search Portals	8
5182	Data processing & Hosting	15
5191	Other Information Services	6

NAICS code	Complementary Industries	Global Market Sales Rank
3231	Printing & Related Support	4
3341	Computer and peripheral equipment	3
3342	Communications equipment	1
3343	Audio and video equipment	2
3344	Semiconductors, other electronic components	5
3345	Navigational, measuring, control instruments	6
3346	Making, reproducing magnetic/optical media	8
3351	Electric lighting equipment	9
3353	Electrical Equipment	7
3359	Other electrical equipment and components	10

Measurement of Copyright Strength and Markets

Index of Copyright Strength

- Park (2005) and Reynolds (2003), updated
- Index covers 120 countries from 1965 – 2013 (annually, where available)
- Four Components (measuring strength of protection):
 - ① Duration (Copyrights and Related Rights)
 - ② Usage (Private Use, Collective Licensing)
 - ③ Enforcement (Injunction, Criminal Penalties, Anti-circumvention)
 - ④ International Agreements/Treaties/Conventions
- Overall Index varies from 0 – 4.
- Source of Information: Statutes, Case Laws, and Commentaries

Copyright Market Size

- Sales of Media Firms, Software companies (total by country x year)
- Source: Thomson Datastream

Trends in Global Copyrights

	Global Copyright Market Size			Index of Copyright Protection		
	1994-1999	2000-2005	2006-2011	1994-1999	2000-2005	2006-2011
Mean	11.6	17.8	30.4	2.45	2.66	2.87
Std Dev	22.6	34.5	51.8	0.65	0.62	0.62
Coef of Variation	1.95	1.94	1.70	0.27	0.23	0.21
Skew	3.94	3.66	3.23	-0.37	-0.59	-0.53
G6/Non-G6	11.9	12.1	8.6	1.32	1.27	1.23
Corr w/ Index of Copyrights	0.32	0.33	0.36	1.00	1.00	1.00

Copyright market size is in billions of real 2005 dollars. Index of copyright protection is on a 0 - 4 scale.

G6/Non-G6 refers to the ratio of the mean among G6 countries to that among non-G6 countries.

Sample statistics are calculated across 70 countries and 18 years.

Descriptive Statistics

Technology Transfer by Copyright Sector: Rank

Sector	NAICS code	FDI Stock	Unaffiliated Licensing	Affiliate R&D
Newspaper, Periodicals, Books ...	5111	7	4	6
Software Publishers	5112	1	1	1
Motion Pictures & Video Industries	5121	6	2	2
Sound Recording	5122	8	6	n/a
Radio, TV, and Cable Networks	5151-5152	3	3	7
Internet, Data Processing	5161, 5181-5182	5	7	5
Telecommunications	5171-5179	2	5	3
Other Information	5191	4	8	4

Sample period: 1994-2011. The numbers represent within-sector rankings of U.S. parent companies' worldwide activities.

Preview

Figure 1: Ratio of Tech Transfer Post-Copyright Reform to Pre Copyright Reform

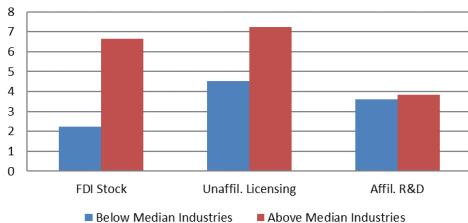
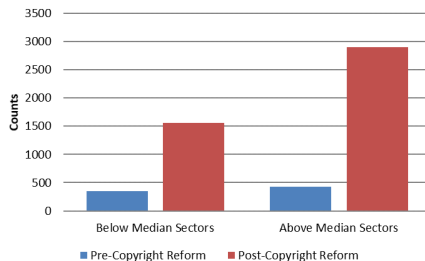


Figure 2: Episodes of FDI Entry and Copyright Reform, by Industry Type



Note: Median is normalized to 1. It is the ratio of the median firm sales of a sector to that of all sectors pooled.

Impacts of Copyrights

A. Copyright Reform and Tech Transfer						
DEP VAR	(1) FDI	(2) FDI	(3) ULic	(4) ULic	(5) Aff R&D	(6) Aff R&D
Copyright Reform	0.947** (0.314)	1.210** (0.579)	0.026 (0.089)	-0.098 (0.215)	-0.099 (0.160)	-0.437** (0.206)
Copyright Reform x Indus Rank		-0.197 (0.339)		0.093 (0.138)		0.253** (0.124)
Control Variables	Included	Included	Included	Included	Included	Included
Observations	7,611	7,611	7,611	7,611	7,611	7,611
Adj R-sq	0.432	0.432	0.774	0.774	0.442	0.443

See Notes below

B. Copyright Intensity and Tech Transfer						
DEP VAR	(1) FDI	(2) FDI	(3) ULic	(4) ULic	(5) Aff R&D	(6) Aff R&D
Copyright Protection	2.263*** (0.749)	2.861** (1.093)	-0.009 (0.347)	-0.732* (0.427)	-0.553 (0.395)	-1.433*** (0.517)
Copyright Protection x Indus Rank		-0.535 (0.701)		0.646*** (0.182)		0.787** (0.359)
Control Variables	Included	Included	Included	Included	Included	Included
Observations	7,611	7,611	7,611	7,611	7,611	7,611
Adj R-sq	0.432	0.432	0.774	0.774	0.443	0.444

Robust standard errors in parentheses, clustered by country

*** p<0.01, ** p<0.05, * p<0.1, Firm, Year, & Country Effects included

Controls: Host GDP, Wages, Tax Rate, Legal Rights Quality, Parent & Affil Sales

Impacts of Copyrights

C. Copyright Reform and Tech Transfer: Excluding G6						
	(1)	(2)	(3)	(4)	(5)	(6)
	Excluding Canada, France, Germany, Italy, Japan, and the U.K.					
DEP VAR	FDI	ULic	Aff R&D	FDI	ULic	Aff R&D
Copyright Reform	1.059*	-0.036	-0.359			
	(0.616)	(0.210)	(0.220)			
Copyright Reform x Indus Rank	-0.063	0.012	0.334***			
	(0.352)	(0.140)	(0.124)			
Copyright Protection				2.594***	-0.772**	-0.781*
				(1.150)	(0.357)	(0.462)
Copyright Protection x Indus Rank				-0.312	0.724***	0.573*
				(0.721)	(0.204)	(0.323)
Control Variables	Included	Included	Included	Included	Included	Included
Observations	4,484	4,484	4,484	4,484	4,484	4,484
Adj R-sq	0.420	0.825	0.492	0.419	0.827	0.491

Robust standard errors in parentheses, clustered by country

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, Firm, Year, & Country Effects included

Controls: Host GDP, Wages, Tax Rate, Legal Rights Quality, Parent & Affil Sales

Impacts of Copyrights

D. Count Data Analysis: Extensive Margin, by Industry				
DEP VAR	(1) FDI Episodes	(2) FDI Episodes	(3) FDI Episodes	(4) FDI Episodes
Sample	Full	Full	Non-G6	Non-G6
Copyright Reform	-0.345** (0.174)		-0.024 (0.156)	
Copyright Reform $\times$ Indus Rank	0.408*** (0.114)		0.194* (0.107)	
Copyright Protection		-0.428** (0.169)		0.027 (0.125)
Copyright Protection $\times$ Indus Rank		0.547*** (0.132)		0.270*** (0.091)
In α	-1.864*** (0.359)	-2.102*** (0.457)	-5.166 (14.47)	-15.487*** (2.143)
Control Variables	Included	Included	Included	Included
Observations	3,724	3,724	2,959	2,959
McFadden's Pseudo R-sq	0.273	0.280	0.236	0.239

Standard errors in parentheses, clustered by country

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$, Industry, Year, & Country Effects included
 Controls: Host GDP, Wages, Tax Rate, Legal Rights Quality, Parent & Affil Sales
 α is the overdispersion parameter

Impacts of Copyrights

E. Test of Endogeneity between Copyright Reform and FDI, Full Sample				
DEP VAR	(1) FDI Stock	(2) FDI Stock	(3) FDI Episodes	(4) FDI Episodes
Copyright Reform (t-4)	0.167	0.860	-0.009	0.106
Copyright Reform (t-3)	-0.196	-0.374	-0.068	-0.161
Copyright Reform (t-2)	0.084	-0.122	0.191	0.278
Copyright Reform	0.599*	1.170**	0.225**	0.463**
Copyright Reform (t+1)	0.468***	0.263	-0.002	-0.017
Copyright Reform (t+2)	0.465***	0.050	0.047	0.198
Copyright Reform (t+3)	0.035	0.127	0.060	-0.206
Copyright Reform (t+4)	0.166	0.517*	0.221**	-0.423***
Interaction Copyright Reform (t-4) & Indus Rank		-0.563		-0.104
Interaction Copyright Reform (t-3) & Indus Rank		0.167		0.086
Interaction Copyright Reform (t-2) & Indus Rank		0.143		-0.072
Interaction Copyright Reform & Indus Rank		-0.434		-0.219
Interaction Copyright Reform (t+1) & Indus Rank		0.136		0.018
Interaction Copyright Reform (t+2) & Indus Rank		0.301**		-0.148
Interaction Copyright Reform (t+3) & Indus Rank		-0.070		0.260*
Interaction Copyright Reform (t+4) & Indus Rank		-0.236		0.551**
Control Variables	Included	Included	Included	Included
Observations	7,611	7,611	3,724	3,724
Adjusted R-sq	0.434	0.435		
McFadden's Pseudo R-sq			0.270	0.281

Standard errors in parentheses, clustered by country. Not reported to conserve space

*** p<0.01, ** p<0.05, * p<0.1, Firm, Year, & Country Effects included

Controls: Host GDP, Wages, Tax Rate, Legal Rights Quality, Parent & Affil Sales

Cross-Market Impacts

Parent Company in the Complementary Industry:		
	Number of Affiliates in Copyright-related Indus.	FDI in Copyright Indus. (relative to pre-reform)
Before Copyright Reform	101	1
Post Copyright Reform	356	2.8
Parent Company in the Copyright-Related Industry:		
	Number of Affiliates in Complementary Indus.	FDI in Complementary Indus. (relative to pre-reform)
Before Copyright Reform	87	1
Post Copyright Reform	188	1.5

Post-reform FDI Stock is normalized by pre-reform values

Parent companies in complementary industries had affiliates in the copyright industry in 41 host countries

Parent companies in the copyright industry had affiliates in the complementary industries in 21 host countries

Cross-Market Impacts

A. Market Effects on Complementary Industries: Full Sample

DEP VAR	(1) FDI	(2) Aff R&D	(3) Aff Capital	(4) ULicen	(5) Aff Licen
Copyright Market Size	-0.145 (0.089)	-0.047 (0.101)	0.126** (0.058)	-0.006 (0.040)	0.015 (0.089)
Copyright Market Size x Indus Rank _g	0.086** (0.032)	0.067*** (0.020)	-0.002 (0.025)	0.055* (0.028)	0.057*** (0.018)
Control Variables	Included	Included	Included	Included	Included
Observations	19,704	19,704	19,704	19,704	19,704
Adj R-sq	0.473	0.385	0.577	0.363	0.335

B. Market Effects on Complementary Industries: Non-G6 Sample

DEP VAR	(1) FDI	(2) Aff R&D	(3) Aff Capital	(4) ULicen	(5) Aff Licen
Copyright Market Size	-0.209* (0.124)	-0.121 (0.095)	0.170** (0.082)	0.015 (0.055)	-0.053 (0.095)
Copyright Market Size x Indus Rank _g	0.128*** (0.047)	0.082* (0.044)	-0.031 (0.033)	0.035 (0.039)	0.084** (0.036)
Control Variables	Included	Included	Included	Included	Included
Observations	11,378	11,378	11,378	11,378	11,378
Adj R-sq	0.499	0.404	0.598	0.347	0.326

Robust standard errors in parentheses, clustered by country

*** p<0.01, ** p<0.05, * p<0.1. Firm, Year, & Country Effects included

Indus Rank_g is based on World Sales to Copyright Industries

Controls: Host GDP, Wages, Tax Rate, Legal Rights Quality, Parent & Affil Sales

Recap

- Copyright-related works comprise a significant share of *knowledge* goods and international technology transfer
- Copyright reforms and protection are positively associated with U.S. multinational tech transfer (FDI, unaffiliated licensing, affiliate R&D) in the *copyright-related industries*
- Copyright reforms and protection affect FDI at the intensive and extensive margins
- Causal direction is most likely from reform to FDI, not the reverse
- Copyright reforms are associated with cross-industry acquisitions and investments (e.g., electronic parent companies invest in software affiliates)
- Market size of copyright-related works is positively associated with tech transfers and investments in *complementary industries* (e.g., computer equipment, electronics, communications)
- Copyright regime (and market size) have greater influences on firms in industries with greater than median U.S. (world) sales
- Effects are significant even when the major G6 host countries are excluded

Extensions

- Future work could examine the role of *limitations and exceptions* to copyrights, such as Fair Use, Fair Dealing
- Most economic studies on copyrights are skewed towards research on “rights”, rather than exceptions
- Exceptions refer to the ability to copy or use copyrighted works without permission (e.g., quotations, parodies, format-shifting, data mining)
- Exceptions $\neq$ Piracy
- *Limitations and exceptions* can also add economic value
 - Reduce transactions costs of licensing, search, monitoring, and obtaining permission
 - Enhance value of the creative work, spur transformative works
 - Alleviate market power
 - Reduce cost of creative activity (esp. in developing world)